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EXECUTIVE SUMMARY OF THE THESIS

Analysis and Design of Advanced Internal Model Control Schemes for Linear Systems

LAUREA MAGISTRALE IN AUTOMATION AND CONTROL ENGINEERING - INGEGNERIA DELL'AUTOMAZIONE

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1. Introduction

This thesis explores the concept of robust output regulation in linear time-invariant systems that are affected by persistent external inputs, such as step functions, sinusoidal waves, harmonics, and periodic signals. The Internal Model Principle claims that effective asymptotic tracking and disturbance rejection can only be achieved when the controller includes a model of the exosystem responsible for generating the exogenous signals [4, 9]. In practice, however, sophisticated internal models—whether multi-harmonic or repetitive—make the controller more complex, potentially complicating the stabilisation phase.

The primary contribution lies in the investigation and application of Compensation of Internal Model (CIM) architecture for output regulation [2, 3, 10]. The approach is introduced for state-feedback regulation and extended to output regulation, with an internal model acting on a subset of measurement channels. Under structural conditions, the closed-loop dependence on internal-model parameters becomes affine. This transformation enables online adaptation to unknown or shifting exosystem parameters while maintaining stability without changing the sta-

biliser [6, 11].

A benchmark for grid-connected inverters is employed for validation. CIM is used to implement sinusoidal tracking and harmonic rejection in proportional-resonant and repetitive current controllers, particularly in scenarios where the grid voltage experiences distortion and frequency variations occur [5].

2. Adaptive CIM for state-feedback

Consider the plant

$$\dot{x}(t) = Ax(t) + Bd(t) + Bu(t),$$

where $x(t) \in \mathbb{R}^n$ is its measurable state, $u \in \mathbb{R}^m$ is a control input, and $d \in \mathbb{R}^m$ is a load disturbance. B has full column rank, and (A, B) is stabilisable. Assume that the whole state is measured and that the disturbance can be represented in the form

$$D(s) = M(s)D_0(s) \quad (1)$$

in the Laplace domain, where $M(s)$ is a known model (exosystem) and $D_0(s) \in H_\infty$, i.e., $D_0(s)$ is proper and stable. Moreover, assume

$$M(\infty) = I \quad \text{and} \quad M^{-1} \in H_\infty.$$

The control law is specified in the form

$$u = M(v + R_s x),$$

where M is incorporated into the controller to asymptotically reject the disturbance d by the Internal Model Principle, $v \in \mathbb{R}^m$ can be introduced as a reference signal. The stabiliser is defined as

$$R_s = \bar{R} - \Pi, \quad (2)$$

$\Pi \in H_\infty$ is any (stable) system such that

$$\Pi(s)(sI - A)^{-1}B = I - M^{-1}(s).$$

The possible choice of $\bar{R}$ is

$$\bar{R} = K,$$

where $K \in \mathbb{R}^{m \times n}$ is such that $A + BK$ is Hurwitz.

The key trick for the adaptation is to use the parametrisation of the model in the form

$$\begin{aligned} M^{-1}(s) &= M_0(s) + \Theta M_1(s) \\ &= \left[\begin{array}{c|c} A_0 & B_0 \\ \hline C_1 & I \end{array} \right] + \Theta \left[\begin{array}{c|c} A_0 & B_0 \\ \hline C_2 & 0 \end{array} \right] \\ &= \mathcal{LFT}_l \left(\left(\left[\begin{array}{c|c|c} A_0 & B_0 & 0 \\ \hline C_0 & I & I \\ \hline C_1 & 0 & 0 \end{array} \right], \Theta \right) \right) \end{aligned} \quad (3)$$

for a real parameter Θ and a Hurwitz A_0 . $\mathcal{LFT}_l$ stands for lower linear-fractional transformation (LFT) [12].

For example, the harmonic disturbance $d = \mu \sin(\omega t + \beta)$ can be represented in the form (1) as [7]

$$M(s) = \frac{(s + a)^2}{s^2 + \omega^2} \quad \text{and} \quad d_0(t) = e^{-at}(\alpha t + \beta)\mu,$$

where $\alpha = \omega \cos \phi - a \sin \phi$ and $\beta = \sin \phi$. Since $M^{-1} \in H_\infty$ (refer to the assumption in [1]), all zeros of $M(s)$ lie in the open left half-plane; thus, $a > 0$. The model's inverse is

$$\begin{aligned} M^{-1}(s) &= \frac{s^2 + \omega^2}{(s + a)^2} = \frac{s^2}{(s + a)^2} + \frac{\omega^2}{(s + a)^2} \\ &= \left[\begin{array}{c|c} A_0 & B_0 \\ \hline C_1 & I \end{array} \right] + \omega^2 \left[\begin{array}{c|c} A_0 & B_0 \\ \hline C_2 & 0 \end{array} \right], \end{aligned}$$

with

$$\begin{aligned} A_0 &= \begin{bmatrix} 0 & 1 \\ -a^2 & -2a \end{bmatrix} & B_0 &= \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ C_1 &= \begin{bmatrix} -a^2 & -2a \end{bmatrix} & C_2 &= \begin{bmatrix} 1 & 0 \end{bmatrix}. \end{aligned}$$

The controller $R : (v, x) \mapsto u$ can be defined as $R(s) = \mathcal{LFT}_l(\Psi, \Theta)$, where

$$\begin{aligned} \Psi(s) &= \\ &= \left[\begin{array}{c|c|c} A_0 - B_0 C_0 & B_0 & B_\Pi \\ \hline -C_0 & I & K + C_0 B_0 B^\# \\ \hline -C_1 & 0 & -C_1 B_0 B^\# \end{array} \middle| \begin{array}{c} -B_0 \\ -I \\ 0 \end{array} \right] \end{aligned}$$

and $B_\Pi = B_0(K + B^\# A) - (A_0 - B_0 C_0)B_0 B^\#$. Replacing Θ with its estimation $\hat{\Theta}$ results in the same scheme; moreover, the closed-loop system is stable if $\hat{\Theta}$ is bounded. The scheme can be depicted as illustrated in Figure 1.

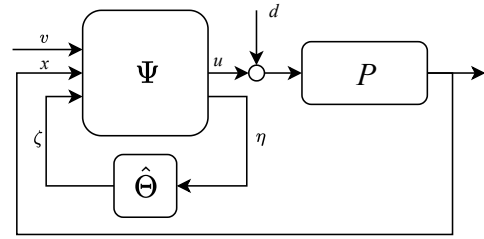


Figure 1: Scheme of the adaptive state-feedback controller.

3. Output regulation with CIM

Consider a continuous-time linear time-invariant (LTI) plant P under a control input $u(t) \in \mathbb{R}^m$ and a measured output $y(t) \in \mathbb{R}^p$. The regulated signal can be defined as

$$y_e(t) = Ey(t) \in \mathbb{R}^{p_e},$$

where $E \in \mathbb{R}^{p_e \times p}$ is assumed to be normalised ($EE' = I$). A possible structure of the controller is [2]

$$R = R_s(E'ME + I - E'E), \quad (4)$$

where $M \in \mathbb{R}^{p_e \times p_e}$ is an internal model whose poles coincide with the poles of the external signal. R_s is a stabiliser of the closed loop designed to meet the performance requirements of the system. Each pole of M should have geometric multiplicity p_e for robust regulation [8]. The stabiliser is designed for the augmented plant

$$P_{aug} = (E'ME + I - E'E)P. \quad (5)$$

Define the matrix $E_\perp \in \mathbb{R}^{(p-p_e) \times p}$ as any matrix that satisfies $E'_\perp E_\perp = I - E'E$. If $p > p_e$, then every such $E_\perp$ has full row rank and satisfies

$E_{\perp} [E' \ E'_{\perp}] = [0 \ I]$. Under the assumptions stated in [2], stabilising P_{aug} can be reduced to stabilising an equivalent plant

$$\bar{P} = (I + E'_{\perp} \Pi_2 E) P_{aug} (I + \Pi_1 E P_{aug})^{-1},$$

obtained by introducing the CIM terms Π_1 and Π_2 through the controller

$$R_s = \bar{R} (I + E'_{\perp} \Pi_2 E) - \Pi_1 E. \quad (6)$$

It is easily apparent that when $y_e = y$, the $E_{\perp}$ matrix is null, and the stabiliser has an identical structure to (2).

Equivalent plant $\bar{P}$ should be such that compensators defined as [2]

$$\begin{bmatrix} \Pi_1 \\ \Pi_2 \end{bmatrix} = \begin{bmatrix} I \\ E_{\perp} \bar{P} \end{bmatrix} (E \bar{P})^{-1} - \begin{bmatrix} I \\ E_{\perp} P \end{bmatrix} (EP)^{-1} M^{-1}$$

are stable. A key result is that $\bar{P}$ can always be chosen to be of the same order as P (refer to [2]). Moreover, if EP is stably invertible and has a proper relative degree (for example, 1 for a model of a harmonic signal), $\bar{P}$ can be chosen as P . Additionally, using an affine parametrisation of M^{-1} analogous to (3), but with the parameter appearing on the right (see [6]), the compensators can be written as

$$\begin{aligned} \begin{bmatrix} \Pi_1 \\ \Pi_2 \end{bmatrix} &= \begin{bmatrix} I \\ E_{\perp} \bar{P} \end{bmatrix} (E \bar{P})^{-1} - \begin{bmatrix} I \\ E_{\perp} P \end{bmatrix} (EP)^{-1} M_0 \\ &\quad - \begin{bmatrix} I \\ E_{\perp} P \end{bmatrix} (EP)^{-1} M_1 \Theta. \end{aligned}$$

Under the assumption that $(EP)^{-1} M_1$ and $E_{\perp} P (EP)^{-1} M_1$ are stable [6], the equivalent plant can be selected independently of Θ so that

$$\begin{aligned} \begin{bmatrix} \hat{\Pi}_1 \\ \hat{\Pi}_2 \end{bmatrix} &= \begin{bmatrix} I \\ E_{\perp} \bar{P} \end{bmatrix} (E \bar{P})^{-1} \\ &\quad - \begin{bmatrix} I \\ E_{\perp} P \end{bmatrix} (EP)^{-1} M_0 \in H_{\infty}. \end{aligned} \quad (7)$$

This also permits replacing Θ with its estimate $\hat{\Theta}$ and obtaining the controller for the original plant as follows:

$$R = \mathcal{LFT}_l(\Psi, \hat{\Theta}),$$

where

$$\Psi = \begin{bmatrix} \Psi_{11} & \Psi_{12} \\ \Psi_{21} & \Psi_{22} \end{bmatrix},$$

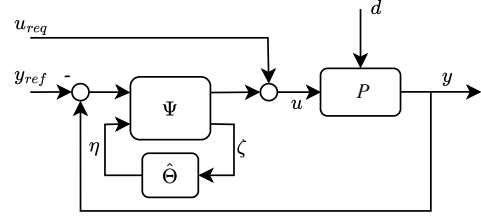


Figure 2: Scheme of the 2DOF controller with CIM and affine model parameters.

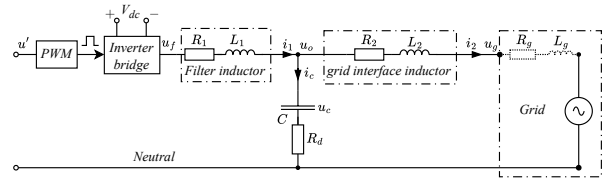


Figure 3: Representation of the plant.

with

$$\begin{aligned} \Psi_{11} &= \tilde{R} + (I - \tilde{R}P)(EP)^{-1}E - (E\tilde{T}_d)^{-1}M_0^{-1}E, \\ \Psi_{12} &= (I - \tilde{R}P)(E\tilde{P})^{-1}M_0^{-1}M_1, \\ \Psi_{21} &= M_0^{-1}E, \\ \Psi_{22} &= -M_0^{-1}M_1, \end{aligned}$$

which provides the basis for the adaptive scheme [6]. The controller scheme is shown in Figure 2.

4. Application: grid-connected inverter

The benchmark is a single-phase grid-connected inverter with an LCL filter, where the grid-side current i_2 is regulated to track a sinusoidal reference in the presence of grid-voltage harmonics and frequency uncertainty. The plant is shown in Figure 3.

The state vector is $x = [i_1 \ i_2 \ u_c]^T$, where i_1 is the inverter-side inductor current, i_2 is the grid-side (injected) current, and u_c is the capacitor voltage. The disturbance input is the grid voltage u_g , the control input is u , and the measured outputs are (i_2, i_1) ; regulation is applied to i_2 ($p_e = 1$), while i_1 is used only for stabilisation/performance shaping and is omitted in the repetitive-control case. Using Kirchhoff's laws, the state-space model is

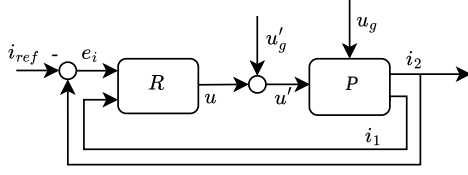


Figure 4: Block diagram of the PR-control with CIM.

$$\begin{aligned}\dot{x} &= Ax + B_1 w + B_2 u \\ y &= Cx + D_1 w + D_2 u,\end{aligned}$$

or

$$P = \left[\begin{array}{c|c|c} A & B_1 & B_2 \\ \hline C & D_1 & D_2 \end{array} \right]$$

$$= \left[\begin{array}{ccc|ccc} -\frac{R_1+R_d}{L_1} & \frac{R_d}{L_1} & -\frac{1}{L_1} & \frac{1}{L_1} & \frac{1}{L_1} & 0 \\ \frac{R_d}{L_{2g}} & -\frac{R_d+R_{2g}}{L_{2g}} & \frac{1}{L_{2g}} & -\frac{1}{L_{2g}} & 0 & 0 \\ \frac{1}{C} & -\frac{1}{C} & 0 & 0 & 0 & 0 \\ \hline 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

5. PR control with CIM

The proportional-resonant (PR) control with CIM (see Figure 4) is examined under grid voltage disturbed by the 3rd and 5th harmonics:

$$u_g = 15 \sin \omega t - 1.5 \sin \omega_2 t - 1.5 \sin \omega_3 t$$

where $\omega_1 = 2\pi \cdot 50$, $\omega_2 = 2\pi \cdot 150$, $\omega_3 = 2\pi \cdot 250$. Thus, the internal model can be chosen as

$$M(s) = \frac{(s+a)^6}{(s^2 + \omega_1^2)(s^2 + \omega_2^2)(s^2 + \omega_3^2)},$$

with $a > 0$. Regulated output applies solely to the grid current i_2 ($p_e = 1$). Consequently, the internal model is only implemented within the i_2 channel. The selection matrix is $E = [1 \ 0]$, and two compensators should be used in the stabiliser (6).

Figure 5 reports simulation results for PR current control using CIM, including time-domain tracking and the steady-state current spectrum under distorted grid voltage. Frequency adaptation is addressed in the thesis and compared for two approaches: the adaptation law in [6] and a phase-locked loop estimate used to retune the internal model.

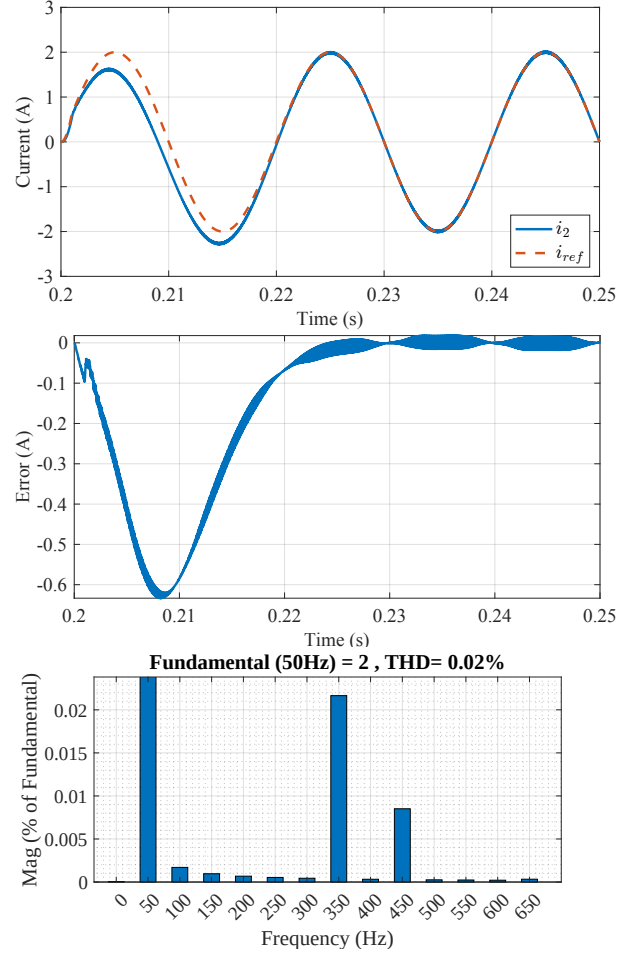


Figure 5: PR control with CIM: grid current tracking with transient (top), tracking error (middle), and output current spectrum (bottom).

6. Repetitive control with CIM

The repetitive control with CIM is examined under (i) grid voltage disturbed by the 3rd and 5th harmonics and (ii) with a connected nonlinear load (an uncontrolled converter loaded with an LC filter and a resistor). In addition, the CIM approach was compared with standard H^∞ repetitive current control [5]. Here, it was considered that only one current i_2 is measured; thus, $E = 1$ and $E_\perp = 0$.

Firstly, define the internal model for the repetitive element as [3]

$$M(s) = \frac{1}{1 - F(s)e^{-\tau s}} I_{p_e},$$

where $F(s)$ is a stable finite-dimensional low-pass filter, $\tau > 0$, and I_{p_e} is a $p_e \times p_e$ identity matrix. In this case, the system model in the state space representation has an internal delay;

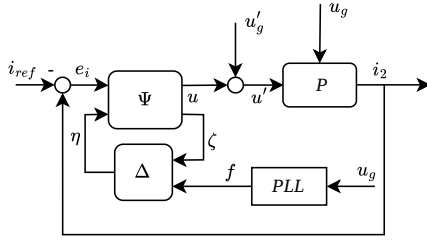


Figure 6: Block diagram of the repetitive current control with CIM and delay adaptation.

however, one can write the inverse of the model based on the (3) as

$$\begin{aligned} M^{-1}(s) &= 1 - F(s)e^{-\tau s} \\ &= M_0(s) + M_1(s)e^{-\tau s}. \end{aligned}$$

Figure 6 represents the scheme with a possible slow delay adaptation mechanism. PLL is used for delay estimation, and the output frequency f is fed to element Δ , where the delay element is calculated as

$$\Delta = e^{-(f^{-1} - w_\phi^{-1})s}.$$

For the benchmark parameters, the channel from u to i_2 is stable and minimum-phase (verified on the nominal model); hence, it is stably invertible. In addition, taking into account that the relative degree of the plant is 2, the equivalent plant can be chosen as $\bar{P} = P$ with the second-order Butterworth filter for the repetitive element.

$$F(s) = \frac{w_\phi^2}{s^2 + \sqrt{2}w_\phi + w_\phi^2},$$

where w_ϕ is a tuning parameter on which tracking and disturbance rejection capability depend. Since $E = 1$ and $E_\perp = 0$, the only compensator Π_1 is used. Moreover, since $\bar{P} = P$, the terms in (7) cancel, yielding $\hat{\Pi}_1 = 0$ (and $\hat{\Pi}_2 = 0$). The remaining compensator term becomes

$$\Pi_1 = -P^{-1}M_1e^{-\tau s}.$$

Regulator (6) for the augmented plant (5) is

$$R_s = \bar{R} - \Pi_1 = \bar{R} + P^{-1}M_1e^{-\tau s}.$$

Therefore, applying the equations above and the M^{-1} definition, the controller (4) is

$$R = R_s M = (\bar{R} + P^{-1}M_1e^{-\tau s})(1 + M_1e^{-\tau s})^{-1}.$$

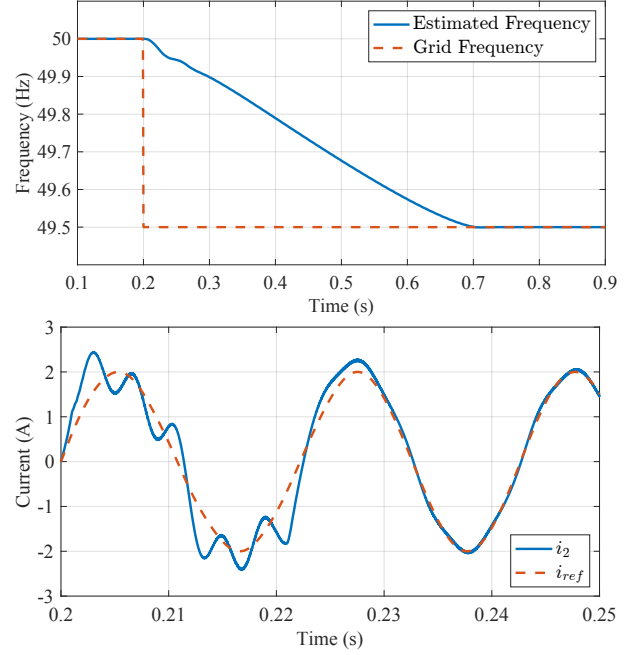


Figure 7: Simulation results for adaptation of current repetitive control with CIM from 50Hz to 49.5Hz.

The stabiliser is selected as a static output-feedback gain $K = -1.1$.

7. Representative results

Figure 7 shows the transient during estimation provided by the PLL after a step change in the grid frequency.

A compact quantitative comparison of H^∞ repetitive control and repetitive control with CIM is reported in Table 1. Robustness is quantified by the minimum Nyquist distance to -1, $d_{\min} = \min_\omega |1 + L(j\omega)|$, where L is the open-loop transfer function; smaller values of $d_{\min}$ correspond to a smaller robustness buffer. The results highlight the expected trade-off: the CIM-based repetitive controller achieves strong harmonic rejection at the cost of a slightly reduced robustness buffer compared with the H^∞ -based stabiliser used as a baseline. This can be mitigated by refining the stabiliser design. In particular, since the achieved THD is already well below the H^∞ baseline, there is margin to retune the stabiliser toward increased robustness while maintaining satisfactory harmonic rejection performance.

	H^∞	CIM
Overshoot, %	< 0.1%	< 0.1%
Settling time, s	0.32	0.02
THD (nonlinear load), %	5.22	1.10
THD (harmonics in grid), %	14.58	1.97
Min. Nyquist distance to -1	0.61	0.53

Table 1: Performance and robustness of the repetitive controllers.

8. Conclusions

CIM provides a structured approach for implementing the Internal Model Principle, separating internal-model embedding from stabiliser synthesis. For state-feedback and output-feedback regulation, including partial regulation, CIM reduces the impact of internal-model complexity on stabiliser synthesis and allows unknown internal-model parameters to be modified without redesigning the stabiliser. A grid-connected inverter benchmark shows that CIM-augmented PR and repetitive controllers achieve good steady-state tracking and harmonic rejection under grid distortion and frequency variations.

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